Midterm - Jordan algebra

October 27nd, 2025

Choose up to five problems from this list to submit.

1. Let A be simple associative algebra. Show that A^+ is simple Jordan algebra.
2. Let A be commutative associative algebra with a derivation D satisfying $D^3 = 0$. Define a new multiplication in A
$$a \star b = D(a)D(b).$$
Show that $(A, \star)$ is a Jordan algebra.
3. Let J be Jordan algebra. Show that for any $a, b \in J$ the operator $[R_a, R_b]$ is a derivation of J.
4. Let J be Jordan algebra and let e be an idempotent in J. Show that U_e is a projection on J_1, where J_1 is the Peirce decomposition component with respect to e.
5. Let J be Jordan algebra with unit element. Define invertible element in J. Prove that a is invertible in J if and only if U_a is invertible in $MultJ$.
6. Let J be a Jordan algebra. Show that if $Mult(J)^n = 0$ then $J^{2n-1} = 0$.
7. Show that Peirce decomposition components satisfy the following relations:
$$J_{ii}^2 \subseteq J_{ii}, \quad J_{ij}^2 \subseteq J_{ii} + J_{jj}, \quad J_{ij}J_{jk} \subseteq J_{ik} \ (i, j, k \text{ distinct}).$$
8. Show that for any finite-dimensional Jordan algebra J its nil-radical is hereditary, namely for any ideal I of J we have $NilI = I \cap NilJ$.
9. Prove that for octonion algebra $\mathbb{O}$ (over real numbers), $\mathbb{O}^+$ is isomorphic to a Jordan algebra of a symmetric nondegenerate bilinear form.